What To Look For In LED BULBS

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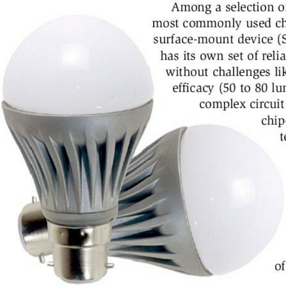
Think of lighting and one cannot ignore LED products, especially LED bulbs which have gained immense popularity in the last few years. And while you may think you know it all while buying an LED bulb, there are many crucial aspects that you may be unaware of. So you end up purchasing poor-quality LED bulbs. Here we give you a lowdown on all the related factors so that you make a well-informed decision before you part ways with your money.

The technical checklist

Lumens, watts, efficacy and power factor are common LED lighting terms. A good-quality LED bulb is one that consumes less wattage of electricity. Lumens, which are a measure of brightness, should be high, and the consequent efficacy, i.e., lumens per watt, should be greater than 100. Power factor indicates the power fraction actually utilised. It should be 0.9 or higher. While you may be aware of these basics, there are a plethora of other factors to consider when buying an LED bulb.

LED chipset. The quality of the chipset used highly influences the performance of any LED product. LED bulbs using a high-quality chipset will provide better output and last longer, but cost more.

Among a selection of chipset types, the most commonly used chipset is based on the surface-mount device (SMD) technology. SMD has its own set of reliability factors, but not without challenges like comparatively lower efficacy (50 to 80 lumens per watt) and complex circuit design. This makes chip-on-board (COB) a better proposition for the chipset.



The circuitry of COB simplifies the installation and space utilisation of components. The manufacturing cost of COB chipsets is 5-10 per cent less than of SMD chipsets, which decreases the cost of the end product. Additionally, COB brings higher efficacy (70 to 110 lumens per watt), thus it is used for high lumen output in LED bulbs. However, one drawback of COB technology is the unavailability of colour changing options, which can be achieved through SMD chipsets with upgradations. But then this will result in an increase in the price of the LED bulb.

The latest to hit the market is multiple chips-on-board. The architecture offers even higher efficiency and colour rendering index (CRI), with better heat management. However, at present, this technology is not used much in the Indian market, except for low-wattage LED bulbs.

Surge protection. Inconsistent voltage supply and frequent surges and spikes may overheat and damage the LED circuitry. Surge protection is a blockage mechanism against a power input over a specific threshold, ensuring voltage safety limit for the device. The bulb itself should have a voltage range within 130V-270V, and an in-built surge protection level of at least 1.5-2kV.

CRI. CRI is a measure of the ability of a lighting equipment to reproduce colours of light as close as possible and is marked between 1 and 100. Higher CRI helps in better colour distinction and visuals while decreasing strain on the eyes. Typically, LED bulbs have a CRI of over 82.

Correlated colour temperature (CCT). It is an identification of the colour of light emitted by an LED equipment. Colours are meas-

Understanding LED bulbs

An LED bulb consists of an LED chipset and a driver, which is the main power management system that drives the input voltage and current to the LED chip in utilisable form. Additionally, there is a heat-sink, which channels the heat generated by the chip during light production away from the internal components. There is also a lens, which directs the emitted light towards the intended area.